

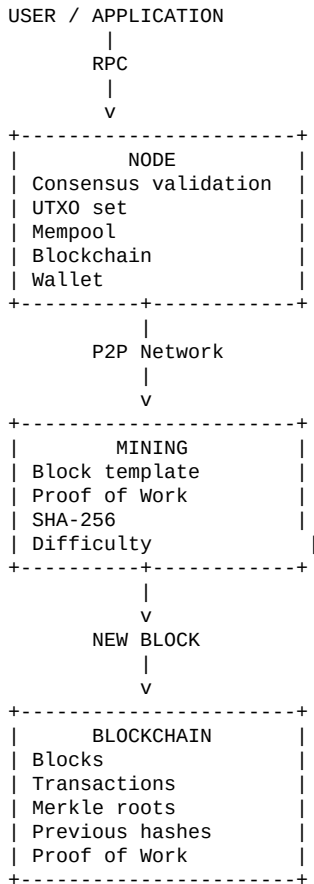
Bitcoin Main Components and QTC Architecture

A component-by-component technical overview

#	Component	Brief explanation
1	1. Blockchain	The blockchain is Bitcoin's permanent public historical record. Blocks are cryptographically linked through previous-block hashes.
2	2. Block Structure	A block packages transactions with consensus data such as the previous block hash, timestamp, difficulty target, nonce, and Merkle root.
3	3. Consensus Mechanism	Bitcoin uses Proof of Work (PoW) to establish which valid chain has the greatest accumulated computational work.
4	4. Mining Algorithm	Bitcoin mining uses double SHA-256 on block-header data. Miners repeatedly change adjustable fields, especially the nonce, until the block hash is valid.
5	5. Mining	Mining creates candidate blocks, performs Proof of Work, broadcasts successful blocks, and earns the block subsidy plus transaction fees.
6	6. Difficulty Adjustment	Bitcoin adjusts mining difficulty every 2,016 blocks so the long-term average block interval targets about 10 minutes.
7	7. Block Reward	A successful miner receives a block subsidy through the coinbase transaction plus transaction fees included in the block.
8	8. Halving	The block subsidy is reduced approximately every 210,000 blocks, roughly once every four years at the intended block interval.
9	9. Transactions	Bitcoin transactions consume previously unspent outputs and create new outputs with spending conditions.
10	10. UTXO System	UTXO means Unspent Transaction Output. Bitcoin tracks spendable outputs rather than maintaining conventional account balances.
11	11. Script	Bitcoin Script defines the conditions that must be satisfied to spend transaction outputs. It supports signatures, multisignature constructions, and conditional logic.
12	12. Cryptography	Bitcoin traditionally uses ECDSA signatures over the secp256k1 elliptic curve for transaction authorization.
13	13. Hashing	SHA-256 is used extensively in Proof of Work, block hashing, Merkle trees, and other protocol operations. RIPEMD-160 is also used for address generation.
14	14. Merkle Tree	Transactions in a block are combined into a Merkle tree, producing a single Merkle root that is committed to by the block header.
15	15. Wallet	A wallet manages private keys and related metadata, derives addresses, creates transactions, selects inputs, calculates fees, and signs transactions.
16	16. Private Keys	Private keys are secret cryptographic values that authorize spending under corresponding spending conditions. Anyone controlling the keys can spend the outputs.
17	17. Public Keys	Public keys are derived from private keys and are used by the network to verify digital signatures.
18	18. Addresses	Addresses are human-friendly representations of Bitcoin spending conditions. Major formats include legacy, P2SH, SegWit, and Taproot.
19	19. Full Node	A full node independently downloads, validates, and relays blocks and transactions according to Bitcoin's consensus rules. It does not mine.
20	20. P2P Network	Bitcoin uses a decentralized peer-to-peer network in which nodes communicate directly with other nodes to exchange transactions, blocks, and network data.
21	21. Mempool	A node's mempool holds valid, unconfirmed transactions that the node is willing to relay and potentially include in a future block.
22	22. Block Propagation	Nodes relay newly received valid blocks across the peer-to-peer network so other nodes can verify and build on them.
23	23. Transaction Propagation	Nodes relay transactions across the network before confirmation, subject to their local validation and relay policies.
24	24. Chain Selection	Bitcoin nodes follow the valid chain with the greatest accumulated Proof of Work, subject to the protocol's consensus rules.
25	25. Confirmations	A transaction receives its first confirmation when included in a valid block. Additional blocks built on top increase its depth in the chain.
26	26. Monetary Policy	Bitcoin's protocol defines a limited issuance schedule with a nominal maximum supply of 21 million BTC, subject to the exact consensus rules.
27	27. Genesis Block	Block 0 is Bitcoin's genesis block and establishes the beginning of the blockchain.
28	28. Coinbase Transaction	The coinbase transaction is the special first transaction in a block that collects the block subsidy and eligible transaction fees for the miner.
29	29. Transaction Fees	Users attach fees to transactions to compensate miners for block-space consumption and to influence transaction inclusion priority.
30	30. Time Mechanism	Block timestamps and difficulty adjustment provide Bitcoin with a protocol-level approximation of time. They are not a centralized clock.
31	31. Network Rules	Consensus rules define what transactions and blocks are valid. Full nodes enforce these rules independently.
32	32. Node Software	Bitcoin Core is the dominant reference implementation, while other compatible implementations also exist. Node software implements the protocol.
33	33. Mining Software	Mining software constructs block templates and coordinates Proof of Work, usually sending work to specialized mining hardware.
34	34. Mining Pools	Mining pools coordinate many miners, distribute work, and generally distribute rewards according to contributed work under the pool's rules.

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35	35. RPC Interface	Bitcoin Core's RPC interface allows applications and scripts to query and control supported node functions programmatically.
36	36. APIs and Explorers	External services expose blockchain data in user-friendly forms. These are application-layer services rather than core consensus components.
37	37. HD Wallets	Hierarchical deterministic wallets derive many keys and addresses from wallet seed material, making backups and key management easier.
38	38. Seed Phrase	A seed phrase is a human-readable representation of wallet seed material used by many wallet systems to recover derived keys.
39	39. Script Types	Bitcoin has multiple transaction/output constructions, including P2PKH, P2SH, SegWit constructions, and Taproot.
40	40. Protocol Upgrades	Bitcoin can evolve through proposals and consensus-compatible upgrades, commonly documented through Bitcoin Improvement Proposals (BIPs).

Bitcoin Architecture in Layers



Transactions and UTXOs

Bitcoin does not fundamentally maintain conventional account balances. Transactions consume UTXOs and create new UTXOs. For example, spending 1.2 BTC might consume 1 BTC + 0.7 BTC and create 1.20 BTC for the recipient, 0.49 BTC as change, and 0.01 BTC as a fee.

Mining

Bitcoin mining uses double SHA-256 Proof of Work. Miners repeatedly alter block-header data until the resulting hash is below the current target.

Proof of Work

Proof of Work makes block production computationally expensive. Nodes select the valid chain with the greatest accumulated work rather than simply trusting a miner.

Difficulty

Difficulty is adjusted every 2,016 blocks to target an average block interval of approximately 10 minutes.

Wallets

Wallets manage seed material, private keys, public keys, addresses, UTXO discovery, transaction construction, fee selection, signing, and backups.

Full Nodes

Full nodes independently verify transactions, UTXOs, signatures, scripts, blocks, Proof of Work, subsidy rules, and other consensus rules.

P2P Network

Bitcoin nodes exchange transactions and blocks directly with peers. There is no central Bitcoin server maintaining the ledger.

Mempool

A node's mempool contains valid, unconfirmed transactions that may later be included in blocks.

Merkle Trees

Transactions are recursively hashed into a Merkle tree. The Merkle root is committed to by the block header.

Bitcoin System Stack



QTC Comparison

Bitcoin component	Bitcoin	QTC direction
Blockchain	Bitcoin blockchain	QTC blockchain
Consensus	Proof of Work	Proof of Work
Mining	SHA-256	RandomX/Cuckoo-oriented design under consideration
Mining hardware	ASIC-dominated	CPU-oriented / ASIC-resistant goal
Signatures	ECDSA / secp256k1	Dilithium 3 in the current QTC design
Key establishment	Classical cryptographic mechanisms	Kyber1024 where applicable
Hashing	SHA-256	SHA3-512 preference / selected PQ-safe primitives
Addresses	bc1... and other formats	qtc1... target format
Transactions	UTXO	UTXO-style design
Node	Bitcoin Core and compatible nodes	QTC node
P2P	Bitcoin P2P	PQ-aware P2P design
Wallet	Bitcoin wallet ecosystem	QTC post-quantum wallet
Block structure	Bitcoin block format	QTC-specific format/rules
Block time	Target ~10 minutes	QTC-selected target
Supply	21 million BTC	84 million QTC design target
Block size	Bitcoin consensus limits	QTC-selected block limit
Mining reward	Bitcoin issuance schedule	QTC issuance schedule
Difficulty	Bitcoin retargeting	QTC retargeting
Script	Bitcoin Script	Compatible or modified QTC scripting
RPC	Bitcoin Core RPC	QTC RPC
Upgrades	BIPs	QTC improvement proposals

QTC Architecture Summary

- Ledger: UTXO-style blockchain.
- Consensus: Proof of Work.

- Mining: CPU-oriented / ASIC-resistant design using the RandomX/Cuckoo direction under development.
- Signatures: Dilithium 3 in the current implementation direction.
- KEM / key establishment: Kyber1024 where applicable.
- Hashing: SHA3-512 preference.
- Addresses: qtc1... target format.
- Networking: QTC P2P with a post-quantum-aware handshake design.
- Wallet: Post-quantum wallet capable of managing QTC keys and transactions.
- Node: QTC node implementing validation, storage, networking, mempool, and RPC.
- Monetary policy: 84 million QTC design target.
- Protocol: QTC-specific block, difficulty, issuance, scripting, and upgrade rules.

Important Distinction

The QTC items in this document describe the project direction discussed for QTC rather than independently verified production implementation. Mining, block parameters, post-quantum networking, and final protocol rules should be treated as specifications until finalized, implemented, tested, and independently reviewed.